

SatQuery AI — Geospatial Analysis Report

Remote Sensing Intelligence & Multimodal Evidence Verification

Job ID:	e654980e-e857-44c4-a6dc-ed1bdcc4724a
Project ID:	e64343dc-4fb3-41c3-ac45-61635ae3beaf
Task Type:	GEOGROUND
Status:	RUNNING
Generated:	2026-09-26 11:38:53 UTC

1. Request Specification

Query / Prompt: "locate circles, squares, buildings"

2. Factual Findings & Observations

Finding 1 (geoground_finding): Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.

Measurements: protocol_version: 1.0, success: True, job_id: 83d749b3-82af-4fd8-b307-87e323fe00be, step_key: geoground_detection, attempt_id: 1, output_assets: {'grounding_overlay': 'C:\\Users\\HP\\earthquery\\storage\\scratch\\83d749b3-82af-4fd8-b307-87e323fe00be\\grounding_overlay.png'}, output_metadata: {'model': 'OWLv2 GeoGround (google/owlv2-base-patch16-ensemble)', 'queries': ['circles', 'squares', 'buildings'], 'total_detections': 2, 'detections': [{'label': 'circles', 'score': 0.3752, 'box': [63.4, 63.5, 194.9, 194.8]}, {'label': 'squares', 'score': 0.2848, 'box': [179.5, 19.9, 241.1, 61.8]}], 'summary': 'Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.', 'text': 'Visual grounding completed with OWLv2: detected 2 object regions matching queries [circles, squares, buildings]. Breakdown: 1 circles, 1 squares.', 'vram_allocated_mb': 1862.86, 'note': 'Open-vocabulary zero-shot visual grounding replacing GeoGround-7B. Operates within 6 GB VRAM limit (~590 MiB allocated).', 'metrics': {'duration_ms': 4538.089752197266, 'peak_vram_mb': 1862.85693359375, 'peak_ram_mb': None, 'device_name': 'cuda', 'cudnn_enabled': False}, error_code: None, error_message: None

3. Operational Boundaries & Limitations

- Observations are generated based on local hardware inference (RTX 5060 Laptop GPU).
- ChangeFormer outputs represent 2D building change masks trained on LEVIR-CD.
- UPerNet semantic segmentation reflects generic ADE20K demo proxy classes unless domain-trained.
- Satellite pixel coordinates are not mapped to geographic bounds unless explicit CRS metadata is verified.
- Analyst confirmation is recommended prior to operational deployment.